Qheadache, Version 1.0

Software Requirements Specification

draft A

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Index

1 Introduction.. 3
1.1 Purpose.. 3
1.2 Scope... 3
1.3 Definitions, Acronyms, and Abbreviations..... 3
1.4 References.. 3
1.5 SRS Document Overview.. 4
2 GENERAL DESCRIPTION 4
2.1 Product Perspective.... 4
2.1.1 User Interfaces 4
2.1.2 Hardware Interfaces 4
2.1.3 Software Interfaces.. 4
2.1.4 Communications Interfaces... 4
2.1.5 Memory Constraints..... 4
2.1.6 Site Adaptation Requirements... 4
2.2 Product Functions.. 5
2.2.1 Undo et Redo Actions.. 5
2.2.2 Time passed to play.. 5
2.2.3 Count of action number.. 5
2.2.4 Score..... 5
2.2.5 Score windows... 5
2.3 User Characteristics.... 5
2.4 Start Up Requirements.... 5
2.5 Apportioning of Requirements.. 5
3 SPECIFIC REQUIREMENTS.... 5
3.1 External Interfaces.. 5
3.1.1 User Interfaces..... 6
3.1.1.1 Introduction.... 6
3.1.1.2 Main window.. 6
3.1.2 Software Interfaces. 6
3.1.3 Communications Interfaces... 6
3.2 Functional Requirements... 6
3.2.1 Actions.. 6
3.2.1.1 Presentation of the board... 6
3.2.1.2 Block selection. 7
3.2.1.3 Block deselection.. 7
3.2.1.4 Block movement. 7
3.2.1.5 Undo Action.... 8
3.2.1.6 Redo Action.. 8
3.2.2 End of the game management.. 9
3.2.2.1 End of the game .. 9
3.2.2.2 Finish Windows with Statistics... 9

3.2.2.3 Simple Finish Windows.....10
3.2.3 Statistics Management.....10
3.2.3.1 Player Statistics management.....10
3.2.3.2 Game Statistics management.....10
3.2.3.3 Statistics erasing.....11
3.2.3.4 Statistic Window.....11
3.2.4 File management.....12
3.2.4.1 Open game.....12
3.2.4.2 Save game.....12
3.2.4.3 Save game as.....13
3.2.4.4 Exit.....13
3.2.5 Menu bar.....14
3.2.5.1 Game menu.....14
3.2.5.2 Action menu.....14
3.2.5.3 Statisctis menu.....15
3.2.5.4 Help menu.....15
3.3 Performance Requirements.....15
3.4 Software System Attributes.....15

1Introduction

1.1Purpose

This specifications establishes the requirements for the product named Qheadache. The intended audience is the analyst, programmer and tester of Qheadache.

1.2Scope

The product is a computerized game that displays an interface used to solve a specific headache.

1.3Definitions, Acronyms, and Abbreviations

Not applicable.

1.4References

Not applicable.

1.5SRS Document Overview

The remaining sections of this document provide a general description, including characteristics of the users of this project, the product' hardware, and the functional and data requirements of the product. General description of the project is discussed in section 2 of this document. Section 2 gives the functional requirements, data requirements and constraints and assumptions made while designing the game. It also gives the user viewpoint of product use. Section 3 gives the specific requirements of the product. Section 3.0 also discusses the external interface requirements and gives detailed description of functional requirements.

2GENERAL DESCRIPTION

2.1Product Perspective

2.1.1User Interfaces

The product runs as a stand-alone application.

Its user interface uses menus, graphics and sounds.

2.1.2Hardware Interfaces

The product requires the use of a keyboard and a mouse to interface with the user. It requires a graphical display of at least 800*600 resolution.

2.1.3Software Interfaces

The product uses the Qt graphical library. It must run with all the operating systems that Qt supports.

2.1.4Communications Interfaces

Not applicable.

2.1.5Memory Constraints

Not applicable.

2.1.6Site Adaptation Requirements

Not applicable.

2.2Product Functions

2.2.1Undo et Redo Actions.

The user must undo and redo its last thousand actions.

2.2.2Time passed to play.

The product must count and display the time that the user uses to play.

2.2.3Count of action number.

The product must count and display the number of the user' action.

2.2.4Score.

The product must record the score (time and number of counts) of a play associated with the name of a user.

2.2.5Score window.

The product must display a window with all the player' scores.

2.3User Characteristics

No qualification is necessary.

2.4Start Up Requirements

Not applicable.

2.5Apportioning of Requirements

Not applicable.

3SPECIFIC REQUIREMENTS

3.1External Interfaces

The product generally requires a mouse and a keyboard for input. Other pointing and input devices are allowable, provided they provide similar functions to a mouse and keyboard, namely the ability to move a cursor onscreen to select buttons and the ability to type names.

The product uses menus, graphics and sounds. The hardware and operating

system must provide an 800x600 screen resolution. Sound is not required to play the game.

3.1.1User Interfaces

3.1.1.1 Introduction

The users consist of anyone who wants to play a simple game who knows how to operate a computer, with a beginning level player starting at age 8, up through an advanced level player who could be an adult.

3.1.1.2Main window

The main window shall provide the following parts :

- A board, see chapter 3.2.1.1.
- a menu bar, see chapter 3.2.5.

3.1.2Software Interfaces

Not applicable.

3.1.3 Communications Interfaces

Not applicable.

3.2Functional Requirements

3.2.1Actions.

3.2.1.1Presentation of the board.

The board is a rectangular zone where the user could move some blocks. Let x be the mesure unit. The height of the board game is 5x, its width is 4x : x can' t be less than 50 pixels and greater than 100 pixels. The blocks are separated by a marge of 0.1x.

There are four square blocks with a side of x.

There are four rectangular blocks with the following dimensions : a height of 2x and a width of x.

There is one block with the following dimensions : a height of x and a width of

2x.

There is one square block with a side of 2x.

The board is black and the blocks are yellow.

3.2.1.2Block selection

3.2.1.2.1Description

See above chapter 3.2.1.1.

3.2.1.2.2Input

Left-clicked down on a block.

3.2.1.2.3Processing

The game state becomes "Block deplacement".

3.2.1.2.4Output

None.

3.2.1.3Block deselection

3.2.1.3.1Description

See above chapter 3.2.1.1.

3.2.1.3.2Input

Left-clicked up on a selected block.

3.2.1.3.3Processing

The game state becomes "Block selection".

3.2.1.3.4Output

None.

3.2.1.4Block movement

3.2.1.4.1Description

See above chapter 3.2.1.1.

3.2.1.4.2Input

Mouse movement during the "Block movement" state.

3.2.1.4.3Processing

The selected block follows the mouse movement without overlapp the other blocks and exit of the game zone. The selected block can't move near other blocks at least 0.05x from the others blocks.

3.2.1.4.4Output

None.

3.2.1.5Undo Action

3.2.1.5.1Descritption

The user can cancel a movement.

3.2.1.5.2Input

Menu selection.

3.2.1.5.3Processing

The game displays the block positions at the places where they were before the last movement. This action is consider like a movement. The "undo" action is unvailable if there was no previous movement.

3.2.1.5.4Output

New game board display.

3.2.1.6Redo Action

3.2.1.6.1Descritption

The user can redo a movement that has been canceled.

3.2.1.6.2Input

Menu selection.

3.2.1.6.3Processing

The game displays the block positions at the places where they were before the last movement was canceled. This action is consider like a movement. The redo action is unvailable, if the previous action wasn't a "undo" action.

3.2.1.6.4Output

New game board display.

3.2.2End of the game management

3.2.2.1End of the game

3.2.2.1.1Description

How the user finishs the game.

3.2.2.1.2Input

The great square is moved at the bottom of the board.

3.2.2.1.3Processing

All the player statistics are freezed.

3.2.2.1.4Outputs

If the number of block movements of the current player is lower than the highest number of block movement recorded in the statistic file, the The "Finish Window with Statistcs" is displayed, see chapter 3.2.2.2. If not the "Simple finish Window' is displayed, see chapter 3.2.2.3.

3.2.2.2Finish Window with Statistics.

3.2.2.2.1Description

The Finish Window with Statisctis contains a the following text : "You win ! Enter your name : ", an Edit Box that can contain 20 characters and a pushbutton with the label "OK".

3.2.2.2.2Input

The games is over, see chapter 3.2.2.1.

3.2.2.2.3Processing

The player clicks on the pushbutton "OK". The player statistics are recorded in the statistic file of the software, according to the requirement of the chapter 3.2.3.2.

3.2.2.2.4Outputs

The "finish" window is closed. The statistic window is displayed, see chapter 3.2.3.4.

3.2.2.3Simple Finish Window

3.2.2.3.1Description

The Simple Finish Window contains a the following text : "You win !" and a pushbutton with the label "OK".

3.2.2.3.2Input

The games is over, see chapter 3.2.2.1.

3.2.2.3.3Processing

The player clicks on the pushbutton "OK".

3.2.2.3.4Outputs

The Simple Finish Window is closed. The statistic window is displayed, see chapter 3.2.3.4.

3.2.3Statistics Management.

3.2.3.1Player Statistics management

3.2.3.1.1Desciption

The following statistics are recorded during the game :

- number of block movements since the start.
- time since the start.

3.2.3.1.2Input

A block movement.

3.2.3.1.3Processing

The number of block movement in incremented of 1. The difference of time of the block movement and the previous recorded time is recorded.

3.2.3.1.4Output

None.

3.2.3.2Game Statistics management

3.2.3.2.1Desciption

The game statistics is composed of 10 player statistics.

3.2.3.2.2Input

The Finish Window with Statistics is completed by the player, see chapter 3.2.2.2.

3.2.3.2.3Processing

The statistics of the player (its name, the block movement number, the time passed to solve the headache) is recorded in the statistic file. If 10 player statistics are already recorded, the player statistics of the file with the greatest number of block movements is erased.

3.2.3.2.4Output

If the file was correctly updated, there is no ouput. If not, like wrong pemissions or disk full, an error message is displayed.

3.2.3.3Statistics erasing

3.2.3.3.1Descritption

The user could erase all the statistics.

3.2.3.3.2Input

Menu selection.

3.2.3.3.3Processing

The data stored in the statistic file are erased.

3.2.3.3.4Output

If the file was correctly updated, there is no ouput. If not, like wrong pemissions or disk full, an error message is displayed.

3.2.3.4Statistic Window.

3.2.3.4.1Descritption

The Player Statistics Window is composed of a listbox of 10 lines. Each line is composed of the name of a player, the number of block movement, the time used by the player to solve the headache.

This statistcs are read from the statistic file of the game.

3.2.3.4.2Input

End of the game (see chapter 3.2.2.1) or menu selection.

3.2.3.4.3Processing

Window display.

3.2.3.4.4Output

None.

3.2.4File management

3.2.4.1Open game

3.2.4.1.1Description

Open a previous saved game.

3.2.4.1.2Input

Menu selection.

3.2.4.1.3Processing

A dialog box is open : the user could choose a file that contains all the data of the game previously saved.

3.2.4.1.4Ouput

The board game is re-draw according to the file data. The player statistics are set to the player statistics of the file data.

3.2.4.2Save game

3.2.4.2.1Description

Save the current game.

3.2.4.2.2Input

Menu selection.

3.2.4.2.3Processing

If the game was never saved, the processing is identiqual to the action "Save as...".

If not, the following internal data are saved into the previous file that was used to to save the the game : the current positions of the blocks, their last 10000 previous positions, the number of the previous movements and the time passed by the user to solve the headache.

3.2.4.2.4Ouput

None.

3.2.4.3Save game as

3.2.4.3.1Description

Save the current game.

3.2.4.3.2Input

Menu selection.

3.2.4.3.3Processing

A dialog box is open : the user could choose a file that will contain all the data of the current game. Next, the following internal data are saved into the file : the current positions of the blocks, their previous positions, the number of the previous movements and the time passed by the user to solve the headache.

3.2.4.3.4Ouput

None.

3.2.4.4Exit

3.2.4.4.1Description

Stop the game

3.2.4.4.2Input

Menu selection.

3.2.4.4.3Processing

If the game is not saved, a dialog box is displayed that asks to the player if he wants to save the game. Two choices are possible : "Yes" and "No". If "Yes" is selected, the action "Save" is processed and the main window disappeared. If "No' is selected, the main window disappeared.

3.2.4.4.4Ouput

None.

3.2.5Menu bar

3.2.5.1Game menu

3.2.5.1.1Description

Contains "Open game ...", "Save Game ...", "Save Game As..." and "Exit". In this order.

3.2.5.1.2Input

Menu selection.

3.2.5.1.3Processing

Action in question is performed :

- "Open game ..." -> action "Open game", see chapter 3.2.4.1.
- "Save Game ..." -> action "Save Game", see chapter 3.2.4.2.
- "Save Game As ..." -> action "Save game as", see chapter 3.2.4.3.
- "Exit" -> action "Exit", see chapter 3.2.4.4.

3.2.5.1.4Output

Menu disappears. Requirements of the action determines the continuation.

3.2.5.2Action menu

3.2.5.2.1Description

Contains "Undo" and "Redo". In this order. The menu selection is unvailable if the associated action is unvailable.

3.2.5.2.2Inputs

Menu selection.

3.2.5.2.3Processing

Action in question is performed :

- "Undo" -> action "Undo", see chapter 3.2.1.5.
- "Redo" -> action "Redo", see chapter 3.2.1.6.

3.2.5.2.4Outputs

Menu disappears. Requirements of the action determines the continuation.

3.2.5.3Statisctis menu

3.2.5.3.1Description

Contains "Display" and "Erase". In this order.

3.2.5.3.2Inputs

Menu selection.

3.2.5.3.3Processing

Action in question is performed :

- "Display" -> displays the statistic window, see chapter 3.2.3.4.
- "Erase" -> erase the statistics, see chapter 3.2.3.3.

3.2.5.3.4Outputs

Menu disappears. Requirements of the action determines the continuation.

3.2.5.4Help menu

3.2.5.4.1Description

Contains "About".

3.2.5.4.2Inputs

Menu selection.

3.2.5.4.3Processing

"About Window" is displayed. The "About Windows" is composed of the following text "Qheadache 1.0 by Jean-Philippe Brossat jp_brossat@yahoo.fr"

3.2.5.4.4Outputs

Menu disappears. Requirements of the action determines the continuation.

3.3Performance Requirements

There can be only one user per machine.

3.4Software System Attributes

The software must be portable to the Windows OS.